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Digital Twin for Gas Condensate Field - Representation of the System from the Edges of Reservoir to Export Point of the CPF

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Abstract

This paper presents the development of a real-time digital twin for a gas-condensate field, covering the full production chain - from the boundaries of multilayer reservoir to the gas processing facility outlet. The objective is to create a continuously operating, ready-to-use model capable of simulating phase behavior, flow dynamics, and compositional changes under varying thermobaric conditions to enhance production forecasting accuracy.

The digital twin architecture enables full-field modeling with automated global and local adaptations. A global adaptation process is implemented across the production history using a maximum likelihood estimation approach grounded in Bayes' theorem. Key subsurface parameters - such as initial reservoir pressure, recoverable reserves, skin factor, threshold pressure, and effective inflow depths - are treated as prior distributions with associated uncertainties. Adaptation is performed by minimizing the total mismatch between model outputs and actual field measurements, including reservoir pressure trends, fluid compositions (C7+ content), individual layer contributions from production logs, and flow rates at varied bottomhole pressures (80%, 100%, and 120% of operational levels).

The posterior estimates of uncertain parameters obtained through Bayesian global adaptation yielded a history-matched model consistent with reverse distribution studies throughout the reservoir. The integrated model accurately replicates the thermobaric and compositional dynamics observed across the production chain, from downhole to processing facility. It successfully captures the effects of reservoir layering and production regime shifts on flow performance. The model remains continuously operational and ready for real-time forecasting and scenario analysis. The integration of diverse data types - reservoir pressures, production logging, and fluid composition - into a single adaptive framework significantly improves forecast reliability and decision-making support across field operations.

This work introduces a novel application of Bayesian global adaptation to a continuously operating digital twin, integrating diverse subsurface and surface data sources. It advances the methodology for real-time, full-system modeling of gas-condensate assets and provides a scalable framework for uncertainty quantification and automated model updating in complex reservoir environments.

Keywords: Digital Twin, PVT modeling, Bayes Theorem, Maximum likelihood, Compositional modeling

Introduction

The economic viability of gas-condensate assets is disproportionately dependent on the efficient recovery of liquid hydrocarbons. The primary threat to this efficiency is retrograde condensation, a phenomenon where declining reservoir pressure below the dew point causes valuable heavy-end components to drop out of the vapor phase and become trapped as an immobile liquid saturation. Furthermore, these assets are often characterized by significant compositional variations between reservoir layers, complex wellbore hydraulics, and surface facilities that are highly sensitive to fluctuating inlet conditions.

Conventional asset management workflows, often relics of a pre-digital era, are ill-equipped for this level of complexity. The inherent latency in this workflow - the time it takes to manually update models and iterate between disciplines - means that by the time a decision is made, the operational reality of the field may have already changed. This fragmented approach inherently fails to capture the real-time, dynamic interactions that govern system performance, leading to suboptimal depletion strategies and reactive, rather than proactive, operational control. The business drivers for a new approach are compelling: market volatility demands greater operational agility, shareholder expectations require maximizing efficiency and value from existing assets, and increasing ESG pressures necessitate better management of emissions and resources.

The Digital Twin concept, when implemented with engineering rigor, provides a solution to these limitations. We define a Digital Twin not as a passive "digital shadow" or a simple monitoring dashboard, but as a living, integrated simulation model of a physical asset, perpetually synchronized with reality through automated data feeds and self-calibration routines. It is a predictive and prescriptive tool, capable of forecasting future performance and recommending optimal operating strategies. It serves as a unified, multi-disciplinary platform - a single source of truth - that enables a holistic understanding of asset behavior.

This paper details the architecture, methodological underpinnings, and operational application of such a Digital Twin for a complex, multi-layer gas-condensate field. Our core objective was to move beyond the static model paradigm and create a self-updating, "evergreen" simulation environment. We will demonstrate that the key enabler for this is a sophisticated, two-tiered adaptation framework. The centerpiece of this framework is a global adaptation engine based on Bayesian inference, a statistically rigorous method for history matching that simultaneously assimilates all available surveillance data. The resulting tool represents a fundamental shift in operational philosophy - from reactive problem-solving to proactive, data-driven optimization and risk management. Figure 1 presents the schematics of Fill cycle modeling implemented in Digital twin of gas condensate field.

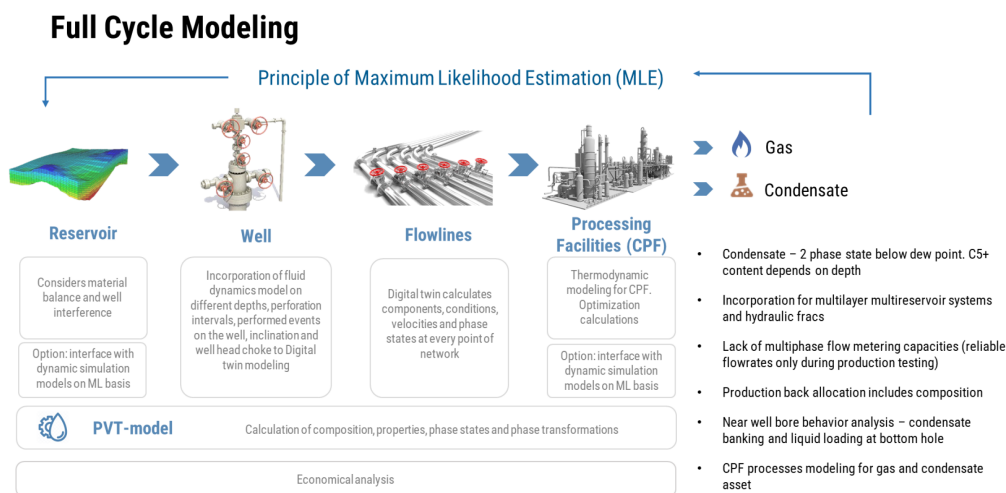


Figure 1—Full Cycle Modeling Schematics

Digital Twin Architecture: The Integrated Asset Model (IAM)

The design philosophy behind the IAM is one of "appropriate fidelity." The goal is not to use the most complex model available for every component, but rather the most appropriate model that balances physical rigor with the computational performance required for real-time applications and large-scale optimization. For instance, while a full-physics numerical reservoir simulator is invaluable for long-term field development planning, its prohibitive runtime makes it unsuitable for the thousands of simulation runs required for an automated optimization routine. Therefore, we employ a carefully selected suite of coupled, physics-based models, each representing a distinct segment of the production chain with the necessary level of detail.

The successful coupling of these models is paramount. The system employs a sequential modular solver, where each component model is solved in turn, from the reservoir to the CPF export, with information passing between them. Iterations are performed until convergence is achieved on key variables like pressure, temperature, and flow rate at the connection points. A critical architectural choice is the use of a single, consistent thermodynamic package (the tuned EOS) across all components. This ensures that a flash calculation performed in the wellbore model is thermodynamically consistent with one performed in the surface separator model, eliminating a common source of error in less integrated systems. Figure 2 presents main calculation modules of Digital twin.

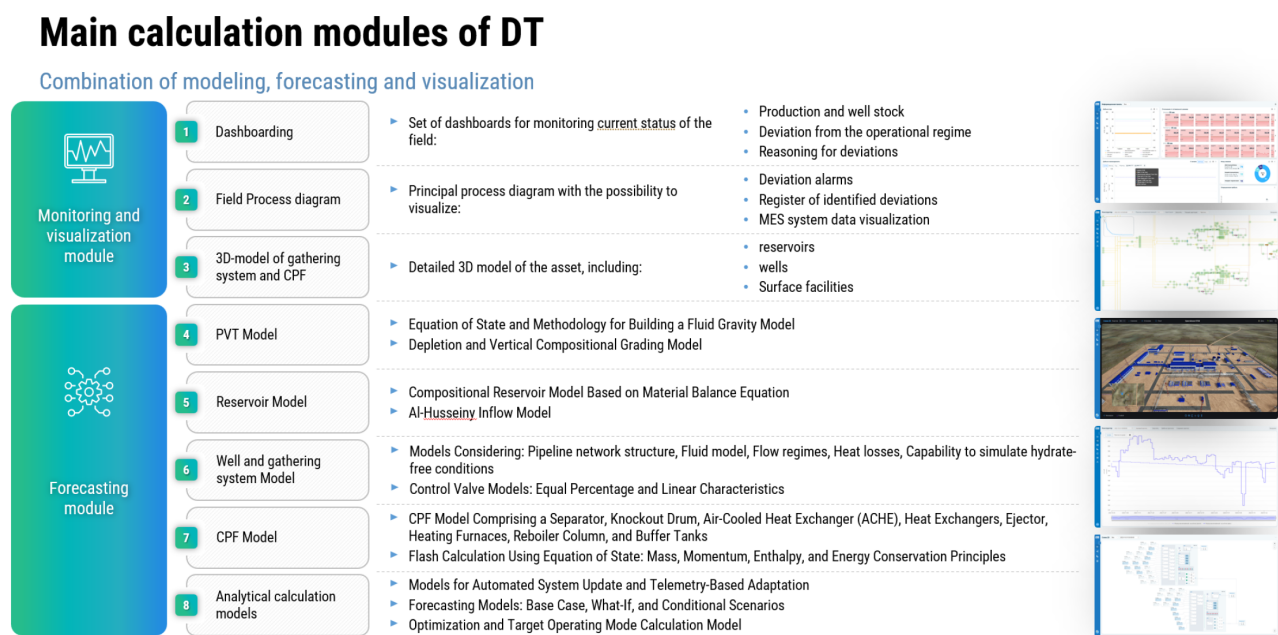


Figure 2—Main calculation modules of DT

PVT and Fluid Characterization

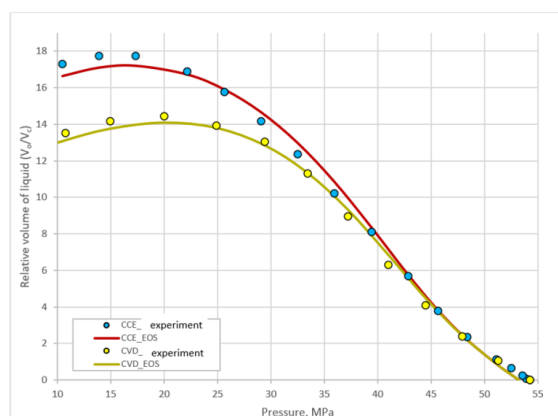
For a gas-condensate system, fluid characterization is the most critical element of the model. An error in the PVT model will propagate throughout every subsequent calculation. Our dual-strategy approach ensures both accuracy and computational efficiency. Figure 3 presents typical achieved results of PVT and CG modeling for PVT and fluid characterization.

- Equation of State (EOS) PVT Model:** A Peng-Robinson (PR) cubic EOS, with volume-shift correction, was selected as the thermodynamic engine. The PR-EOS offers a robust balance between accuracy for both vapor and liquid phase properties and computational speed, which is critical for real-time applications. The EOS was meticulously tuned to match extensive laboratory data, including constant composition expansion (CCE), constant volume depletion (CVD), and

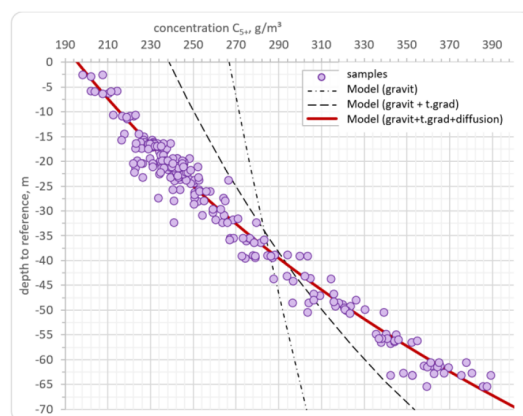
multi-stage separator tests. The tuning process focused on regressing critical properties and binary interaction coefficients of the heavy- end pseudo-components (e.g., C7+) to accurately match the measured dew point, liquid dropout curves, and gas compressibility factors. This detailed tuning is essential for correctly predicting the onset and severity of retrograde condensation.

- Compositional Gradient (CG) Model:** The field's stacked reservoirs exhibit notable variations in fluid composition with depth, likely due to filling history and gravitational segregation. A simple assumption of a uniform fluid composition would lead to significant errors in material balance and production forecasting. The CG model initializes the simulation by establishing the fluid composition (and thus the dew point pressure) as a function of depth for each distinct reservoir unit. This ensures that the commingled fluid entering a wellbore is a correctly flow-weighted average of the initial compositions from each producing layer, a critical detail for accurately modeling the evolution of the wellhead fluid over time.

PVT MODELING APPROACH



CCE and CVD EOS and experiment matching
21 component



Importance of using of gravitational, thermogravimetric and
diffusion models

Figure 3—PVT Modeling approach for PVT and Fluid characterization

Reservoir Inflow Performance Model

The interface between the reservoir and wellbore is modeled using a semi-analytical inflow performance relationship (IPR). The **Al- Hussainy pseudo-pressure formulation** was employed, as it rigorously linearizes the gas diffusivity equation for real gases by integrating pressure-dependent properties (pgZ) over the pressure range. The IPR is defined by:

$$q_g = C \left(p_r^2 - p_{wf}^2 \right)^n,$$

where C is the productivity index coefficient and n is the non-Darcy flow exponent (typically between 0.5 and 1). The model is further enhanced to capture complex near-wellbore effects:

- Composite Skin Factor (S):** This term is not a single fitting parameter but a composite of physical effects: $S = S_d + S_p + Dq_g$. S_d represents formation damage or stimulation from drilling and completion fluids. S_p accounts for the pseudo-skin from partial penetration or restricted entry. The Dq_g term represents the rate-dependent, non-Darcy (or turbulent) flow effect, which becomes significant at high flow velocities near the wellbore.
- Condensate Banking:** This is the most critical IPR effect in a depleting gas-condensate well. As the bottom-hole flowing pressure (p_w) drops below the dew point, a region of high liquid saturation

(the condensate bank) develops around the well, drastically reducing the relative permeability to gas. This phenomenon is modeled by incorporating a set of two-phase pseudo-relative permeability curves into the pseudo-pressure calculation. This makes the IPR dynamic; the well's productivity is not constant but is a strong function of the drawdown and the fluid's phase envelope.

Wellbore and Tubing Hydraulics Model

To accurately model the multiphase flow from the sandface to the wellhead, the **Gray correlation** was implemented. While other correlations exist, the Gray correlation was specifically developed for and is widely validated in high-velocity wet gas and gas- condensate wells. It solves the coupled momentum and energy balance equations along the wellbore, segmented into discrete calculation nodes.

- **Momentum Balance:** This calculates the total pressure drop, accounting for the gravitational head of the gas-liquid mixture, frictional losses against the tubing wall (using a Moody friction factor), and accelerational pressure changes.
- **Energy Balance:** The temperature profile is computed by rigorously modeling heat transfer between the wellbore fluid and the surrounding formation's geothermal gradient. This calculation includes the critical Joule-Thomson effect (cooling of gas upon expansion) and accounts for convective and conductive heat transfer. An accurate temperature profile is not only vital for predicting the potential for hydrate formation but also for correctly evaluating fluid properties (density, viscosity) at each point in the well, which in turn affects the pressure drop calculation.

Surface Network and Process Facility Model

The model's scope extends from the wellhead through the gathering network to the individual components of the CPF.

- **Surface Gathering System:** The network of flowlines and trunklines is modeled using robust multiphase flow correlations (e.g., Beggs and Brill, TUFFP) to accurately predict pressure drops and liquid holdup profiles. This is essential for understanding system backpressure and its impact on well deliverability.
- **Central Processing Facility (CPF):** The CPF is modeled with high fidelity. Instead of a simplified "black box" with fixed efficiencies, each major unit operation is represented by a rigorous, physics-based sub-model. For example, a two-phase separator is not just a splitter; it is modeled as a VLE flash calculation using the full EOS PVT model, determining the precise composition and quantities of the exiting vapor and liquid streams based on the vessel's pressure and temperature. Heat exchangers are modeled using LMTD or NTU methods, with the overall heat transfer coefficient (U-value) treated as a dynamic parameter subject to degradation. Control valves are modeled using the ISA S75.01 standard, calculating pressure drop as a function of the flow coefficient (Cv) and valve opening. This granular approach is indispensable for diagnosing bottlenecks, evaluating debottlenecking options, and optimizing plant performance.

The "Evergreen" Mandate: Automated Model Updating

A static IAM, however detailed, suffers from "model drift" - the inevitable divergence of its predictions from reality as the physical asset ages, degrades, and is operated under new conditions. This drift erodes user trust and ultimately renders the model obsolete. The central tenet of our Digital Twin philosophy is that the model must be "evergreen," perpetually synchronized with the asset. This is not a one-time calibration event but a continuous process, achieved through a structured, two-tiered automated workflow.

Tier 1: Automated Data Actualization

This is the continuous, high-frequency synchronization of the model's boundary conditions with live data from the field's SCADA and data historian systems. Automated data pipelines stream the latest measurements for operational setpoints - wellhead choke positions, separator pressures, compressor speeds, valve statuses - to the Digital Twin platform. This process, running on a cycle of minutes, ensures the twin is always simulating the asset's current operational configuration, making it a true real-time tool.

Tier 2: Automated Model Adaptation

While actualization updates the model's inputs, adaptation updates the model's intrinsic characterization parameters to ensure its outputs match the asset's measured behavior. This is a more profound reconciliation process, subdivided into two distinct but synergistic workflows designed to address different timescales and types of uncertainty.

Global Adaptation: A Bayesian Approach to History Matching

Global adaptation is the cornerstone of the twin's long-term predictive power. It is an automated history-matching process that stands in stark contrast to traditional manual methods, which are often subjective, time-consuming, and non-repeatable. A manual history match performed by two different engineers will almost certainly yield two different models. The Bayesian framework provides an objective, auditable, and repeatable methodology for determining the most probable set of foundational subsurface parameters that collectively explain the entire historical dataset.

The Principle of Holistic Data Integration

Conventional history matching is often a sequential, piecemeal process, leading to non-unique solutions. Our approach is holistic. We construct a single, unified objective function that incorporates every significant piece of historical data. The strength of this method is that it forces a solution that is consistent with all available evidence concurrently, dramatically reducing the non-uniqueness problem. The integrated dataset includes:

- **Static Reservoir Data:** Petrophysical interpretations (porosity, net pay) from well logs.
- **Dynamic Pressure Data:** Pressure transient data from well tests (PBU, DDT) and long-term shut-in surveys, which provide information on permeability-thickness (kh) and reservoir boundaries.
- **Dynamic Fluid Data:** Time-series of produced fluid compositions from gas chromatography, particularly the C7+ fraction, which serves as a powerful natural tracer for tracking depletion and compositional changes.
- **Dynamic Flow Allocation Data:** Inflow contributions from individual layers, as measured by production logging tools (PLTs), which constrain the vertical permeability distribution.
- **System-Wide Telemetry:** The full historical record of pressures and temperatures at key nodes (bottom-hole, wellhead, separator inlets).
- **Facility Production Data:** Historical daily, monthly, and cumulative allocated production of gas, condensate, and water from the CPF.

Mathematical Framework: Bayesian Inference and Maximum Likelihood

The mathematical engine for this global adaptation is rooted in **Bayes' theorem**, providing a formal, probabilistic framework for parameter estimation.

1. **Define Prior Distributions:** We formally acknowledge our initial uncertainty by defining key subsurface parameters not as single deterministic values, but as **prior probability distributions**. For example, the initial gas-in-place (GIIP) for a reservoir layer is defined as a log-normal distribution with

a mean and variance derived from initial volumetric estimates. Parameters treated this way include: initial pressure per layer, GIP, layer-wise permeability, skin factors, and reservoir geometry. These priors act as a form of geologically-sound regularization, preventing the optimization algorithm from converging on physically implausible solutions.

2. **Construct the Likelihood Function:** The likelihood function, $P(\text{dobs}|\text{m})$, quantifies the probability of observing the measured data (dobs) given a particular set of model parameters (m). It is formulated as an objective function that minimizes the weighted sum of squared errors between the model's predictions and the historical measurements across all data types:

$$J(m) = \sum_{i=1}^{\infty} (g(m) - d_{obs})^T * C_d^{-1} * (g(m) - d_{obs})$$

Here, $g(m)$ is the model's forward prediction, and C_d is the data covariance matrix. The diagonal elements of C_d represent the variance (or uncertainty) of each measurement, allowing us to assign higher weight to more reliable data (e.g., a calibrated downhole gauge from a PBU test) and lower weight to less certain data (e.g., daily allocated production).

3. **Solve for the Posterior Distribution:** Bayes' theorem states that the **posterior probability** of the model parameters is proportional to the product of the likelihood and the prior probability:

$$P(m|d_{obs}) = \frac{P(d_{obs}|m) * P(m)}{P(d_{obs})}$$

We seek the **Maximum a Posteriori (MAP)** estimate, which is the parameter set m that maximizes this posterior probability. This is a high-dimensional, non-linear optimization problem, solved using advanced algorithms (e.g., gradient-based methods like L-BFGS or derivative-free methods like evolutionary algorithms). The resulting MAP estimate represents the single best-fit model, rigorously history-matched against all available data.

Local Adaptation: Real-Time Tuning for Operational Fidelity

While global adaptation anchors the model in long-term reservoir physics, local adaptation is a higher-frequency, agile process designed to track the shorter-term evolution of the production system. Its primary focus is on adjusting performance parameters of surface equipment, which can change due to fouling, wear, or minor operational adjustments. This process runs on a daily or even hourly cycle, in contrast to global adaptation, which might be executed on a quarterly basis or after a major field event like a workover or the drilling of a new well.

This process can be framed as a state-estimation problem, where the "state" includes parameters like equipment efficiency or fouling factors. When a persistent mismatch between the twin's real-time predictions and live telemetry is detected, an automated routine adjusts a corresponding performance parameter. For instance, if a heat exchanger's measured outlet temperature consistently deviates from the model's prediction, a recursive algorithm (akin to a Kalman filter) will adjust the model's overall heat transfer coefficient (U-value) to reflect fouling, until the mismatch is resolved. The key challenge is to distinguish true performance degradation from sensor noise or transient fluctuations. This is achieved by implementing smoothing algorithms and strict constraints on the rate and magnitude of parameter changes, which prevents overfitting and preserves the model's predictive integrity.

A critical feature of this system is the use of "constrained adaptation." The algorithms are designed to prevent overfitting; for example, a local adaptation cannot adjust a well's skin factor to explain a short-term production drop if that change would violate the globally adapted reservoir model's pressure response. This hierarchical constraint ensures that short-term tuning does not corrupt the long-term predictive integrity of the model.

Operational Workflow and Applications

The fully operationalized Digital Twin is not a standalone analytical tool but an integrated component of the asset management workflow, enabling a suite of advanced applications.

- **"What-If" Scenario Analysis:** Engineers can perform near-instantaneous simulations to evaluate the system-wide consequences of operational decisions. For example, *"What is the projected impact on CPF inlet separator liquid loading and total condensate yield if we perform an acid stimulation on well-A, assuming a 50% reduction in its skin factor?"*
- **Automated Production Optimization:** The twin is coupled to a non-linear optimization engine. The user defines a business objective (e.g., maximize total condensate revenue) and a set of constraints (e.g., maximum erosional velocity in pipelines, minimum wellhead pressure to avoid liquid loading, CPF capacity limits). The optimizer then automatically runs hundreds of simulation cases to find the optimal combination of controllable variables (e.g., wellhead choke settings) that achieves the objective without violating constraints. The value proposition is a direct and measurable increase in revenue or reduction in OPEX, achieved by operating the complex, interacting system closer to its true optimum on a continuous basis.
- **Proactive Monitoring and Anomaly Detection:** By continuously comparing model predictions against live data, the system serves as a powerful anomaly detection engine. A deviation of measured bottom-hole pressure from the IPR-predicted value can provide an early warning of liquid loading or scale buildup days or weeks before it would be apparent from production trends alone. The value here is in the reduction of unplanned downtime and the enhancement of operational safety and integrity by identifying potential hazards before they escalate.
- **Predictive Maintenance:** The trends in equipment performance parameters, as tracked by the local adaptation module, provide direct, quantitative input for predictive maintenance. For example, the steady decline of a compressor's isentropic efficiency can be extrapolated to predict when it will fail to meet a required discharge pressure, allowing maintenance to be scheduled proactively to avoid a costly unplanned shutdown. This shifts the maintenance strategy from a costly reactive or conservative time-based schedule to a more efficient and cost-effective condition-based approach.
- **Probabilistic Forecasting:** A significant advantage of the Bayesian framework is that it provides a basis for robust uncertainty quantification. By sampling from the posterior parameter distributions (using techniques like Markov Chain Monte Carlo), we can generate a full probabilistic production forecast (e.g., P10, P50, P90 profiles), providing decision-makers with a clear understanding of the range of possible outcomes and the associated risks for long-term strategic planning.
- **Life-of-Field Strategy Development:** The evergreen nature of the twin makes it an invaluable tool for strategic planning. As the model is continuously updated with the latest performance data, it provides the most current and reliable basis for re-evaluating long-term field development strategies. Questions such as the optimal timing for installing additional compression, the economic viability of an infill drilling campaign, or the projected timing of the field's economic limit can be revisited regularly, ensuring that strategic capital decisions are always based on the most up-to-date understanding of the reservoir's capabilities.

Results and Discussion

The implementation of the Digital Twin on the target gas-condensate field yielded substantial results. The global adaptation process successfully converged to a single, consistent set of subsurface parameters that honored more than a decade of diverse surveillance data. For instance, the model successfully captured the abrupt change in the GOR trend of well B-3 following the water breakthrough in an adjacent well, a complex inter-well interference effect that previous siloed models failed to predict. The posterior distributions for

key reservoir parameters, such as the permeability of specific layers and the initial gas-in-place, were significantly "tighter" (i.e., had lower variance) than the initial priors, quantitatively demonstrating the value of information gained from the integrated history match. History-matching plots for well pressures, GOR trends, and produced fluid compositions showed excellent agreement. Critically, the model was able to replicate complex, historically observed behaviors, such as the gradual shift in produced fluid composition as high-permeability, leaner-gas layers depleted faster than lower-permeability, richer-gas layers. The resulting history match demonstrated excellent agreement across all key metrics.

Operationally, the twin proved to be a high-impact decision-support tool. An optimization study identified a new field-wide choke configuration that increased total condensate production by 3% while simultaneously reducing fuel gas consumption in the CPF - a multi-objective improvement that would have been impossible to find through manual, trial-and-error methods. The anomaly detection system successfully flagged the early stages of hydrate formation in a subsea flowline during a cold spell, enabling timely intervention with chemical injection that averted a full blockage.

The primary challenges encountered were twofold: historical data quality and computational demand. Data quality issues, such as sensor drift, missing data, and inconsistencies in early manual production records, required the development of sophisticated data pre-processing and validation algorithms. The computational cost of the global adaptation, which involves thousands of full-field simulations, was addressed by leveraging a scalable, cloud-based high-performance computing (HPC) architecture, which reduced the runtime for a full adaptation from several days to under six hours.

The project also provided significant lessons learned, extending beyond the purely technical domain. A key takeaway was the critical importance of organizational change management. The introduction of a single, integrated tool required breaking down traditional silos between reservoir, production, and facilities engineering teams. Fostering a collaborative environment where all disciplines trusted and utilized the same model was as crucial to the project's success as the sophistication of the underlying algorithms. Early and continuous engagement with end-users, transparent communication about the model's capabilities and limitations, and a dedicated training program were essential for achieving widespread adoption and ensuring the twin became an integral part of the daily workflow rather than a niche analytical tool.

Conclusions

This work has detailed the successful development, implementation, and application of a comprehensive, evergreen Digital Twin for a complex gas-condensate asset. The following conclusions are drawn:

1. The integrated, physics-based modeling approach, coupling the reservoir, wells, and surface facilities, provides a holistic and superior representation of the production system compared to traditional, siloed workflows.
2. A two-tiered automated adaptation framework, combining long-term Bayesian global adaptation with short-term local tuning, is a robust and effective methodology for ensuring the Digital Twin remains perpetually synchronized with the physical asset, effectively eliminating "model drift".
3. The application of Bayesian inference for global history matching is a powerful technique. It rigorously assimilates all available data, honors geological uncertainty through the use of priors, and yields a well-constrained model with formally quantified parameter uncertainty.
4. The operationalized Digital Twin is a transformative tool that shifts asset management from a reactive, problem-solving mode to a proactive, predictive, and optimized paradigm, unlocking significant and measurable value in production, operational efficiency, and safety.
5. The true value of such a system is realized not just through technical excellence, but through successful integration into organizational workflows, fostering cross-disciplinary collaboration and trust in a single source of truth.

6. Looking forward, this platform provides a foundation for further innovation. Future work will focus on integrating machine learning-based proxy models to accelerate optimization runtimes further and extending the model's scope to include reliability and maintenance logistics for a full techno-economic optimization of the asset.
7. Ultimately, the architecture and methodologies presented herein offer a proven, scalable, and repeatable framework for the deployment of high-fidelity Digital Twins across other complex and high-value upstream assets.

Nomenclature

CPF	– Central Processing Facility
EOS	– Equation of State
IAM	– Integrated Asset Model
IPR	– Inflow Performance Relationship
MAP	– Maximum a Posteriori
MLE	– Maximum Likelihood Estimation
PLT	– Production Logging Tool
PVT	– Pressure, Volume, Temperature
VLE	– Vapor-Liquid Equilibrium

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